Zero-Curation SAR Ship Classification and Re-Identification via Multi-Signal AIS Label Resolution and SAR-Native Backbone

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***Abstract*—Building labeled SAR ship datasets has historically required significant manual effort: prior work such as OpenSARShip manually verified each AIS-to-SAR match against the MarineTraffic commercial registry. We present a fully automatic pipeline that eliminates this bottleneck. A four-signal label resolver—combining AIS type codes, ship length, Global Fishing Watch (GFW) classification, and vessel name keywords— resolves 87.3% of 2,088 vessels to reliable type labels without any manual annotation, applied to free Sentinel-1 IW GRD imagery over the Gulf of Mexico paired with free NOAA AIS data. We compare three classification backbones on identically matched chip sets: DINOv2 (optical pretrained ViT-S/14), ConvNeXt-Small (CNN baseline), and CROMA (SAR-native ViT-Base pretrained on Sentinel-1/Sentinel-2 pairs). CROMA achieves the best performance at 83.8% overall accuracy and 76.3% macro-F1 on a three-class problem (Tanker, Cargo, Fishing), outperforming DINOv2 (77.0% OA / 67.6% F1) and ConvNeXt (81.2% OA / 66.8% F1). As a second experiment demonstrating re-identification capability, we apply CROMA with ArcFace metric learning to tanker re-identification across multiple passes: the same vessel imaged on different dates. Rank-1 accuracy of 5.1% on 49 test identities represents 2.5× above random chance and establishes a meaningful baseline for this novel task. Our pipeline, labels, and dataset of 18,342 GRD chips from 2,088 MMSI-labeled vessels are publicly available at https://doi.org/10.5281/zenodo.21000650.**

***Index Terms*—synthetic aperture radar, ship classification, Sentinel-1, AIS, label resolution, CROMA, backbone comparison, ship re-identification, maritime domain awareness**

I. Introduction

Automatic ship type classification from Synthetic Aperture Radar (SAR) imagery supports maritime domain awareness tasks including sanctions enforcement, fisheries monitoring, and port state control [2], [3]. SAR is particularly valuable for maritime surveillance because it operates day and night, penetrates cloud cover, and cannot be evaded by vessel operators the way optical sensors can. The Automatic Identification System (AIS) provides vessel position and identity broadcasts that complement SAR imagery: when AIS is active, it enables tracking of individual vessels across multiple satellite passes. However, an estimated 10–20% of global cargo tonnage operates with AIS disabled or spoofed at any given time—so-called "dark vessels" engaged in sanctions evasion, illegal fishing, or illicit cargo transfer. A classifier trained on labeled SAR chips can potentially identify vessel type even in the absence of AIS, filling a critical gap in maritime monitoring. Yet two barriers limit the scalability and reproducibility of existing classification pipelines.

Manual dataset construction. The dominant labeling strategy—used in OpenSARShip [1], FUSAR-Ship [3], and most published datasets—matches SAR detections to AIS broadcasts and then manually verifies each label through a commercial registry such as MarineTraffic. This process is labor-intensive, region-specific, and cannot be replicated without commercial data subscriptions. It also introduces a selection bias: scenes are chosen near major ports where ship density justifies the curation effort, leaving open-water regions and small-traffic areas unrepresented.

AIS noise. Even with manual verification, raw AIS vessel type codes are unreliable [19]: operators frequently transmit code 0 ("not specified"), misuse categories, or leave type fields unchanged across vessel reflagging. Training directly on AIS codes without noise mitigation produces significantly degraded classifiers.

We address both problems with a free, fully automatic pipeline applied to the Gulf of Mexico—an open-water region free from the dense port clutter that complicates Chinese and European coastal datasets. The pipeline uses only free data sources: Sentinel-1 GRD imagery from the Copernicus programme and NOAA AIS broadcasts.

We make three contributions. (1) A **multi-signal AIS label resolver** that combines four independent signals into a confidence-weighted vote, resolving 87.3% of vessels to reliable labels without manual annotation (Section IV). (2) A **systematic backbone comparison** contrasting an optically pretrained ViT (DINOv2), a CNN baseline (ConvNeXt), and a SAR-native ViT (CROMA) under identical experimental conditions, finding that SAR-domain pretraining provides a consistent advantage (Section V). (3) A **SAR ship re-identification baseline** using CROMA and ArcFace metric learning—identifying whether a given vessel has been previously observed—with MMSI as ground-truth identity, establishing a novel benchmark for this task (Section VI).

II. Related Work

SAR ship datasets. OpenSARShip [1] provides 11,346 Sentinel-1 GRD chips from Chinese port scenes, labeled

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through AIS matching with manual MarineTraffic verification; their k-NN baseline achieves 75–79% OA on three classes (Tanker, Container, Bulk Carrier). FUSAR-Ship [3] extends this with Gaofen-3 high-resolution imagery. SAR-Ship [17] contributes 43,819 Sentinel-1 and Gaofen-3 chips spanning nine vessel categories from Chinese coastal waters, while HRSID [18] provides 5,604 high-resolution Sentinel-1 and TerraSAR-X chips for instance segmentation. All of these datasets rely on manual curation steps that are not reproducible from freely available data alone.

Ship type classification with DL. A comprehensive survey [4] of 74 studies using OpenSARShip and related datasets shows that DL methods achieve 72–93% OA on 3-class ship classification, with the majority in the 77–89% range. The best reported result (93.3% OA) combines AIS with CNN features on a manually curated dataset [8]. Our pipeline achieves 83.8% OA with zero manual curation—competitive with the published range.

SAR-pretrained foundations. CROMA [6] is a contrastive masked autoencoder [12] pretrained on Sentinel-1 GRD paired with Sentinel-2 optical imagery. Its SAR encoder is a ViT-Base [11] producing 768-dimensional embeddings—the only publicly available foundation model pretrained natively on C-band SAR data at this scale. SSL4EO-S12 [16] is a complementary multi-temporal Sentinel-1/2 pre-training dataset, but targets scene-level tasks rather than vessel-level object classification. Prior ship classification work relied on optical backbones (DINOv2, ResNet [9], VGG) applied to pseudo-color SAR renders, introducing a domain mismatch that our CROMA experiments quantify.

Ship re-identification. Person and vehicle re-ID in optical imagery is well studied, with deep metric learning methods [20] consistently benefiting from angular margin losses such as ArcFace [7]. SAR ship re-ID has received no published attention: existing work applies inverse SAR (ISAR) for target recognition in controlled scenarios, while tracking commercial vessels across multiple satellite passes at C-band GRD resolution is an open problem. We establish the first baseline for this task using MMSI as ground-truth identity.

III. Dataset and Pipeline

A. Data Sources

We acquired 241 Sentinel-1 IW GRD scenes [13] over the Gulf of Mexico spanning August–December 2024 from the ASF DAAC (free, Earthdata login). GRD IW mode provides 10 m pixel spacing in VV and VH polarisations, with effective resolution $\sim$20 m $\times$ 22 m.

NOAA Class A and Class B AIS broadcasts for the Gulf of Mexico were downloaded for the same period from the NOAA AIS data portal (free, daily CSV files). These broadcasts include MMSI, vessel type code, ship length, vessel name, and position at the time of transmission.

B. AIS Matching and Chip Extraction

For each scene, AIS messages within a $\pm$10 minute window of the SAR acquisition time were matched to the scene footprint. Ship pixel coordinates were computed by fitting a degree-2 polynomial to the Ground Control Points embedded in the GRD metadata, mapping AIS (longitude, latitude) to image (column, row) with sub-pixel accuracy.

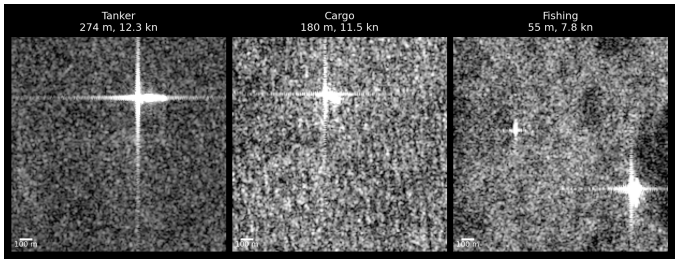


Fig. 1. Representative Sentinel-1 IW GRD VV chips (200×200 px centre crop of the 512×512 chip; scale bar = 100 m). *Left:* 274 m Tanker underway at 12.3 kn — strong cross-shaped response from the large metal hull. *Centre:* 180 m Cargo vessel at 11.5 kn — smaller but distinct point-scatterer signature. *Right:* 55 m Fishing vessel at 7.8 kn in open Gulf water — weaker return consistent with smaller cross-section. All chips are from HIGH-confidence resolved labels.

Dual-polarisation chips of 512×512 pixels centred on each AIS position were extracted and saved as float32 arrays (VV and VH channels). Ships shorter than 100 m were retained for re-ID experiments (minimum 2 passes required); for classification, no length filter was applied beyond a 30-chip minimum per class.

This process produced **18,342 chips** from **2,088 unique vessels**. Of these, 1,942 vessels (93%) appear in two or more passes, enabling within-identity train/test splits for re-identification. Figure 1 shows representative VV-polarisation chips for each of the three classification classes.

C. Dataset Statistics

Figure 2 summarises the vessel distribution. Tanker dominates both HIGH and MEDIUM confidence tiers (544 and 502 vessels), while Fishing is severely under-represented (14 HIGH, 18 MEDIUM). Tug/Special vessels receive no HIGH-confidence assignments because their AIS codes (31, 32, 50–59) rarely co-occur with length or GFW signals that would provide a confirming vote. The length histogram reveals the physical basis for classifier difficulty: Tanker and Cargo vessel lengths overlap substantially in the 100–200 m range, while Fishing vessels are concentrated below 80 m. The 200 m threshold used by the length resolver is visible as a natural gap in the Tanker distribution.

D. Input Representation

For optical-domain backbones (DINOv2, ConvNeXt), chips are converted to a three-channel pseudo-color image: channel 1 VV in dB, channel 2 VH in dB, channel 3 the VV$-$VH difference, each normalised to $[0, 1]$ via the 2nd–98th percentile of finite pixels. The CROMA backbone accepts the two raw channels directly via its own preprocessing (log-scale normalisation to $[0, 1]$ per channel).

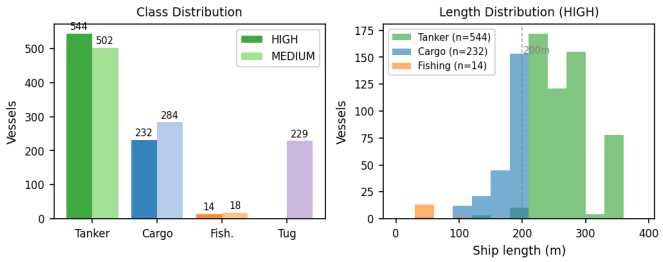


Fig. 2. Dataset statistics. *Left:* Vessel counts per class, split by resolver confidence. Tug/Special receives no HIGH-confidence assignments (all MEDIUM). *Right:* Ship length distribution for HIGH-confidence vessels (3 classification classes). Tanker and Cargo overlap in the 100–200 m range; the dashed line marks the 200 m length-resolver threshold.

IV. Multi-Signal Label Resolution

A. Motivation

Raw AIS vessel type codes suffer from widespread miscoding. To quantify the impact, we trained a DINOv2 classifier with raw AIS codes as labels on the same matched chip set. Overall accuracy reaches 77.5%, but macro-F1 collapses to 44.1%: the model predicts Tanker for virtually every chip (Fishing recall 0%), exploiting class imbalance rather than learning discriminative features. After resolver-based label cleaning, macro-F1 rises to 76.3%—a 32.2 pp improvement—while Fishing recall recovers to 85.7%. The OA gap (77.5% $\rightarrow$ 83.8%) understates the damage: the resolver prevents classifier collapse, not merely label noise.

B. Resolver Design

For each unique MMSI we aggregate four independent signals.

Signal 1: AIS type code. The modal AIS type code across all observations is mapped to a class: 80–89 $\rightarrow$ Tanker, 70–79 $\rightarrow$ Cargo, $\{31, 32, 50\text{–}59\}$ $\rightarrow$ Tug/Special, 30 $\rightarrow$ Fishing. Codes outside these ranges cast no vote.

Signal 2: Ship length. Median AIS-reported length provides signal at the extremes only: $\geq$200 m $\rightarrow$ Tanker (no other class reaches this in Gulf traffic), $<$50 m $\rightarrow$ Tug/Special. The 50–200 m range is ambiguous and casts no vote.

Signal 3: GFW classification. We query the Global Fishing Watch [14] Vessel Identity API for each MMSI. The inferred vessel type (TANKER, CARGO, FISHING, etc.) is mapped to our four classes where possible; OTHER and missing entries cast no vote.

Signal 4: Vessel name keywords. The AIS or GFW vessel name is matched against curated keyword lists: tanker keywords (TANKER, CRUDE, LNG, VLCC, FPSO, CHEMICAL), cargo keywords (BULK, CONTAINER, CARGO, CEMENT, GRAIN), tug keywords (TUG, AHTS, PSV, SUPPLY, OFFSHORE), fishing keywords (FISHING, TRAWL, SEINER).

C. Confidence Voting

Each of the four signals independently casts a vote for a vessel class, or abstains if it has no applicable information. Votes are tallied across signals: three or more agreeing on the same class—or two agreeing with no dissenter—yields HIGH confidence. Two agreeing signals with at least one dissenter, or a single uncontested signal, yields MEDIUM. Fewer than two non-conflicting signals yield UNRESOLVED.

Table I illustrates all four outcomes on real vessels drawn from our dataset. COBIA LNG is the ideal case: AIS type, physical length, and vessel name all independently identify it as a Tanker. THOR INFINITY shows a clean two-signal agreement between AIS type code and GFW classification; length falls in the ambiguous 50–200 m range and correctly abstains. GOLDEN FROST demonstrates graceful conflict handling: its 229 m length exceeds the 200 m Tanker threshold, but AIS type and GFW both say Cargo, so the resolver assigns Cargo at MEDIUM confidence rather than promoting the minority vote. DOLE MAYA illustrates the single-signal MEDIUM case: AIS code 90 ("Other type") maps to no class, length is in the ambiguous range, and only the GFW classification provides a vote. NANCY PETERKIN exposes a genuine limitation: AIS code 52 classifies the vessel as an articulated tug-barge, yet its 206 m length far exceeds any plausible tug size, suggesting miscoding. With one vote for each of two incompatible classes, the resolver correctly withholds a label rather than guessing.

Applied to 2,088 vessels, the resolver achieves **87.3%** resolution; Table II details per-signal coverage and per-class outcomes. HIGH confidence (790 vessels) requires $\geq$3 agreeing votes, or 2 uncontested; MEDIUM (1,033 vessels) requires 2 votes with at most one dissenter; 265 vessels remain Unresolved.

D. Signal Ablation

Table III quantifies each signal's contribution by removing it from the resolver and re-computing resolution outcomes. Removing the AIS type code causes the largest drop: resolution falls from 87.3% to 45.6% ($-$41.3 pp) and HIGH-confidence assignments collapse from 790 to just 18, confirming that AIS codes—despite their noise—remain the primary driver. GFW classification is the most valuable secondary signal ($-$133 resolved, $-$233 HIGH without it). Vessel name keywords contribute marginally because only 0.9% of vessels have discriminative names. Ship length presents a subtle trade-off: removing it slightly *increases* the overall resolution rate (86.9% $\rightarrow$ 91.7%) because length is a noisy signal that creates conflicts when it disagrees with other votes, pushing some MEDIUM-confidence vessels to Unresolved. However, removing length cuts HIGH-confidence assignments nearly in half (790 $\rightarrow$ 402), since it contributes a decisive confirming vote for large Tankers. We retain all four signals in the final resolver because HIGH-confidence labels are the inputs to classification training.

The three-class experiment (Tanker / Cargo / Fishing) uses only HIGH-confidence labels because Tug/Special vessels receive no HIGH-confidence assignments (all 229 require MEDIUM confidence, producing too few chips for a reliable split).

TABLE I
WORKED RESOLVER EXAMPLES FROM THE DATASET. EACH SIGNAL COLUMN SHOWS THE RAW VALUE AND ITS RESULTING VOTE (→ CLASS, OR — IF THE SIGNAL ABSTAINS). THE FOUR ROWS ILLUSTRATE HIGH (3-VOTE AND 2-VOTE UNCONTESTED), MEDIUM (CONFLICTED 2-VOTE AND SINGLE-VOTE), AND UNRESOLVED OUTCOMES.

Vessel	AIS code	Length (m)	GFW type	Name keyword	Vote tally	Conf./Label
COBIA LNG	84 → Tanker	299 → Tanker	OTHER → —	"LNG" → Tanker	3× Tanker	HIGH / Tanker
THOR INFINITY	70 → Cargo	190 → —	CARGO → Cargo	—	2× Cargo	HIGH / Cargo
GOLDEN FROST	70 → Cargo	229 → Tanker	CARGO → Cargo	—	2× Cargo, 1× Tanker	MEDIUM / Cargo
DOLE MAYA	90 → —	195 → —	CARGO → Cargo	—	1× Cargo	MEDIUM / Cargo
NANCY PETERKIN	52 → Tug	206 → Tanker	OTHER → —	—	1× Tug, 1× Tanker	UNRESOLVED / —

TABLE II
AIS LABEL RESOLVER STATISTICS FOR THE 2,088 VESSELS IN OUR DATASET. *Coverage*: VESSELS FOR WHICH THE SIGNAL CAST A NON-NULL VOTE. *Decisive*: VESSELS WHERE THAT SIGNAL CONTRIBUTED TO THE WINNING LABEL.

Signal	Coverage	Decisive
AIS type code	1,911 (91.5%)	1,911 (91.5%)
Ship length	839 (40.2%)	580 (27.8%)
GFW classification	426 (20.4%)	400 (19.2%)
Vessel name	19 (0.9%)	19 (0.9%)

Confidence	Count	Tanker	Cargo	Fishing	Tug/Spcl
HIGH	790	544	232	14	0
MEDIUM	1,033	502	284	18	229
Unresolved	265	–	–	–	–
Resolved	1,823	1,046	516	32	229

TABLE III
RESOLVER SIGNAL ABLATION OVER ALL 2,088 VESSELS. EACH ROW REMOVES ONE SIGNAL; ALL OTHERS REMAIN ACTIVE. HIGH* COUNTS ARE CRITICAL FOR CLASSIFICATION: ONLY HIGH-CONFIDENCE LABELS ENTER THE CLASSIFICATION TRAINING SET.

Condition	HIGH*	MED	Unres.	Res. %	ΔRes.
Full resolver	790	1,033	265	87.3%	—
Remove AIS type	18	935	1,135	45.6%	−870
Remove GFW	557	1,133	398	80.9%	−133
Remove length	402	1,512	174	91.7%	+91[a]
Remove name kw.	783	1,031	274	86.9%	−9

[a]Length creates conflicts; removing it resolves some borderline vessels but halves HIGH-confidence count.

TABLE IV
BACKBONE COMPARISON — 3-CLASS, MATCHED CHIPS (1,283 TOTAL). PER-CLASS VALUES ARE F1 SCORES. AIS-ONLY ROW USES RAW TYPE CODES WITHOUT RESOLVER; ALL OTHER ROWS USE RESOLVED HIGH-CONFIDENCE LABELS.

Backbone	OA (%)	Macro-F1 (%)	Tanker	Cargo	Fishing
DINOv2 (AIS-only)	77.5	44.1	92.9%	39.5%	0.0%
ResNet-50	83.2	74.4	95.7%	41.9%	85.7%
DINOv2 ViT-S/14	77.0	67.6	87.2%	44.2%	71.4%
ConvNeXt-Small	81.2	66.8	96.5%	32.6%	71.4%
CROMA	**83.8**	**76.3**	**94.3%**	**48.8%**	**85.7%**

V. CLASSIFICATION EXPERIMENTS

A. Experimental Setup

All three backbones are evaluated on the same chip set: HIGH-confidence labels, three classes (Tanker / Cargo / Fishing), with per-class chip counts capped at the minority-class limit of 944/287/52 chips (total 1,283). This *matched* condition ensures that backbone differences are not confounded by training data volume. The 70/15/15 stratified split is by class; all chips from a given MMSI appear in exactly one split to prevent leakage.

Training uses AdamW with a backbone learning rate of 10^{-5} and a head rate of 10^{-4}, cosine annealing over 30 epochs, batch size 64. Augmentation: random horizontal/vertical flips and 180° rotation.

ResNet-50: Residual CNN [9] pretrained on ImageNet; ∼25M parameters; fully fine-tuned with a linear classification head.

DINOv2 ViT-S/14: Pretrained on optical imagery [5]; 384-dim CLS token; last 4 transformer blocks unfrozen; linear classification head.

ConvNeXt-Small: Modernized CNN [10] pretrained on ImageNet; ∼50M parameters; full fine-tuning.

CROMA: SAR-native ViT-Base [11] pretrained on Sentinel-1 GRD + Sentinel-2 paired data via masked autoencoding [6], [12]; 768-dim global average pooling of the SAR encoder; last 4 transformer blocks unfrozen; linear classification head.

B. Results

Table IV reports test-set performance. CROMA outperforms both optical-domain baselines across all metrics.

Figure 3 shows validation accuracy and macro-F1 across 30 training epochs. CROMA converges more smoothly than the optical backbones, which exhibit larger epoch-to-epoch fluctuations—consistent with the domain mismatch between their ImageNet pretraining statistics and SAR backscatter. ResNet-50, despite being the smallest model, achieves competitive accuracy through full fine-tuning; DINOv2 shows slower convergence, reflecting the larger distributional shift from optical to SAR inputs.

C. Comparison with Published Results

Table V places our results in the context of published OpenSARShip benchmarks. Direct comparison is complicated by differences in class definitions (OpenSARShip uses Container/Bulk/Tanker; we use Cargo/Tanker/Fishing) and dataset geography (Chinese ports vs. Gulf of Mexico open water).

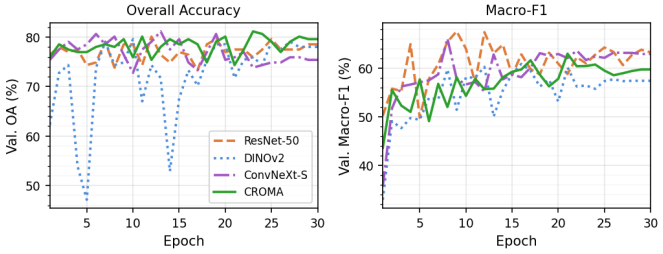


Fig. 3. Validation overall accuracy (*left*) and macro-F1 (*right*) across 30 training epochs for all four backbones. CROMA (solid green) converges most smoothly; optical-domain backbones show higher epoch-to-epoch variance due to domain mismatch with SAR backscatter statistics.

TABLE V
COMPARISON WITH REPRESENTATIVE PUBLISHED RESULTS ON SAR SHIP CLASSIFICATION. OUR DATASET USES A DIFFERENT CLASS SPLIT (CARGO/TANKER/FISHING) AND GEOGRAPHIC REGION (GULF OF MEXICO) FROM OPENSARSHIP BENCHMARKS.

Dataset	Method	OA (%)	Classes
OpenSARShip [1]	k-NN baseline	75–79	3
OpenSARShip [4]	DL methods (range)	77–93	3
OpenSARShip [8]	CNN + AIS fusion	93.3	3
Ours (Gulf)	CROMA (zero-curation)	**83.8**	3

Nevertheless, our zero-curation pipeline achieves 83.8% OA—within the 77–93% range reported across 74 studies in [4] for manually curated datasets—while requiring no manual annotation.

D. Confusion Analysis

The dominant error across all backbones is Cargo predicted as Tanker. Figure 4 shows the CROMA per-class confusion matrix on the test set: 22 of 43 Cargo chips (51%) are misclassified as Tanker, while Tanker recall is 94.3% and Fishing recall is 85.7%. Figure 5 further shows CROMA GRD embeddings in UMAP projection: Cargo and Tanker feature vectors substantially overlap, while Fishing forms two tight, well-separated clusters. This is consistent with the physical reality: both Cargo and Tanker vessels at 10 m GRD resolution present as elongated point-scatterer arrays, and the AIS "Cargo" category (codes 70–79) encompasses visually diverse sub-types (container, bulk carrier, general cargo) that are individually closer to Tankers than to each other.

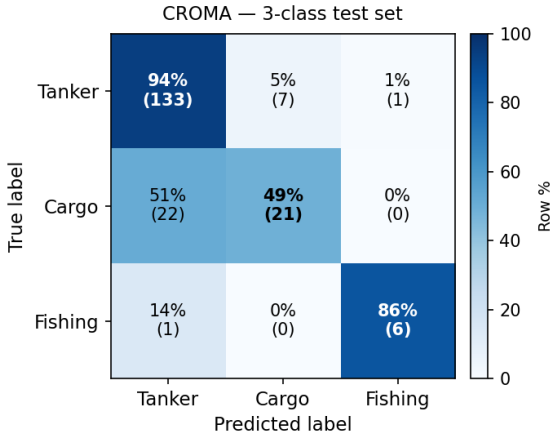


Fig. 4. Per-class confusion matrix for CROMA on the 3-class test set (191 chips). Row-normalised percentages with raw counts in parentheses. The dominant error is Cargo → Tanker (51%): both vessel types present as elongated point-scatterer arrays at 10 m GRD resolution.

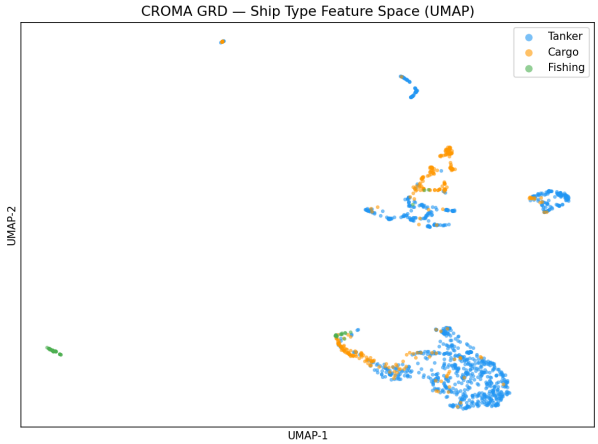


Fig. 5. UMAP [15] projection of CROMA GRD embeddings (test set). Fishing vessels form tight isolated clusters (two orientation modes); Cargo substantially overlaps Tanker, explaining the dominant classification error.

VI. SHIP RE-IDENTIFICATION

A. Task Definition

Re-identification asks: *given a SAR chip of an unknown vessel, retrieve the same vessel from a gallery of previously imaged ships.* This differs from classification, which assigns a type label. Re-ID uses MMSI as ground-truth identity—no manual labeling is required beyond the AIS match. The application is dark-vessel tracking: a vessel that disables its AIS transponder can still be re-identified across passes by its SAR signature.

B. Method

We apply CROMA with ArcFace metric learning [7] ($s = 64$, $m = 0.5$) to tanker vessels only (HIGH-confidence, $\geq$100 m length, $\geq$2 passes). The metric learning head projects the 768-dim CROMA embedding to a 128-dim hypersphere; ArcFace trains angular margin separation between identities. Train/val/test split is by vessel: 235 training ships / 49 test

TABLE VI
TANKER RE-IDENTIFICATION WITH CROMA + ARCFACE (49 TEST SHIPS)

	Rank-1 (%)	Rank-5 (%)	mAP (%)
Random chance	2.0	10.2	–
CROMA + ArcFace	**5.1**	**14.0**	**12.3**

ships (no MMSI appears in more than one split). Standard re-ID metrics are reported: Rank-1, Rank-5, and mean Average Precision (mAP), evaluated as nearest-neighbor retrieval in embedding space.

C. Results

Rank-1 accuracy of 5.1% represents 2.5× above random chance (1/49 ≈ 2%) on a 49-identity closed-set gallery. The modest absolute value reflects the inherent difficulty: tankers at 10 m GRD resolution are visually similar within class, and each vessel has on average fewer than 5 passes available for training (∼915 chips / 235 ships ≈ 3.9 chips/ship). Rank-5 at 14.0% (1.4× random) and mAP at 12.3% confirm a learnable identity signal exists in CROMA embeddings beyond random retrieval.

D. Embedding Analysis

Figure 6 projects the re-ID embeddings into two dimensions via UMAP, colored first by vessel type and then by individual MMSI identity. The type projection (left) shows clear macro-structure: Tanker/LNG vessels cluster at the top, Tug/Towing vessels form several dense satellite groups, and Cargo occupies a broad central region. The network is not simply memorizing noise—it has learned physically meaningful representations of vessel class and size from Sentinel-1 backscatter. Yet the identity projection (right) of the same embedding space reveals no within-cluster separation: every cluster is a rainbow of distinct MMSIs. The model cannot tell *which* tanker from *that there is a tanker*. At 10 m GRD resolution, individual hull geometry and superstructure detail—the cues that would separate two 250 m crude carriers—fall below the resolution floor. This establishes both the feasibility bound of GRD re-ID and the motivation for sub-meter SAR as future work.

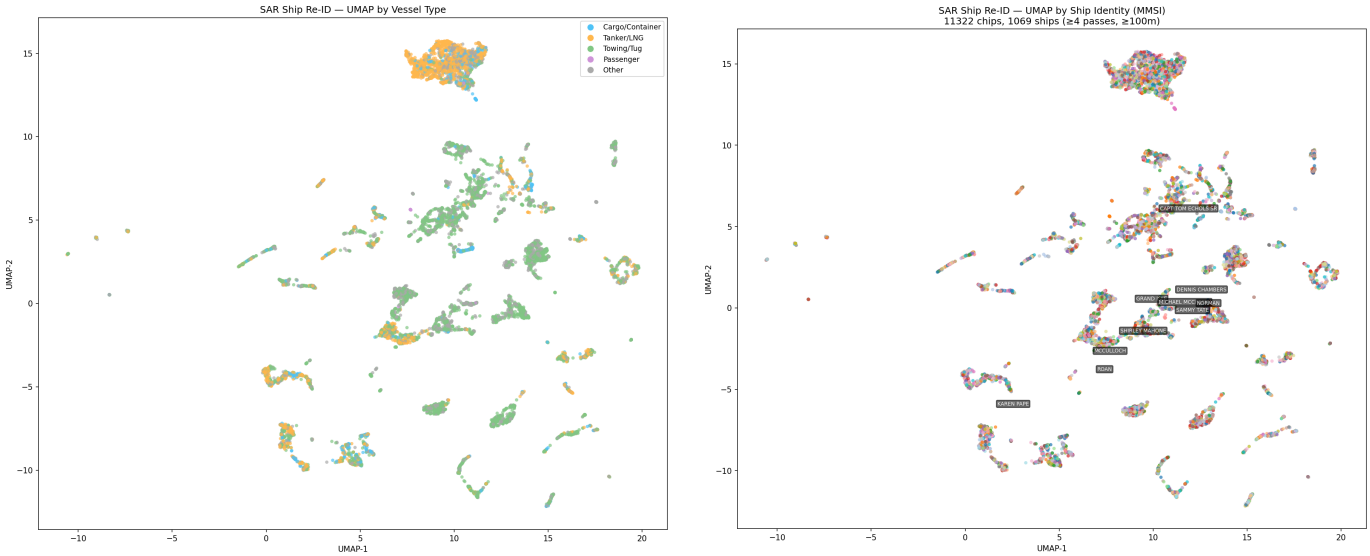


Fig. 6. UMAP [15] of CROMA re-ID embeddings (11,322 chips, 1,069 vessels, $\geq$4 passes and $\geq$100 m length), colored by vessel type (*left*) and by MMSI identity (*right*). The type projection reveals clear macro-structure at 10 m GRD resolution: Tanker/LNG, Tug, and Cargo clusters emerge and separate. The identity projection of the same space shows no within-cluster separation—each cluster is a mixture of distinct identities (colors). Individual hull signatures fall below the GRD resolution floor, establishing the performance ceiling for identity-level retrieval on Sentinel-1 data.

VII. Discussion

Why CROMA outperforms optical backbones. DINOv2 and ConvNeXt were pretrained on natural images, where color texture and photometric gradients carry most discriminative information. SAR backscatter is determined by target geometry, surface roughness, and dielectric properties—physically distinct from optical image statistics. CROMA's pretraining on paired Sentinel-1 and Sentinel-2 data teaches the SAR encoder to extract scattering-mechanism features rather than optical-analogue textures, explaining its advantage even when applied through a linear head. The training curves (Figure 3) reinforce this interpretation: CROMA converges more smoothly than optical backbones, which exhibit larger validation variance due to domain mismatch. Despite this, ResNet-50 matches CROMA's OA (83.2% vs. 83.8%), suggesting that CNN inductive biases partially compensate for domain mismatch when fully fine-tuned; the gap is clearest in macro-F1 (+1.9 pp for CROMA), where Fishing recall is the deciding factor.

Cargo confusion as an inherent limit. The AIS class "Cargo" (codes 70–79) is heterogeneous: it covers Container Ships, Bulk Carriers, General Cargo, and Roll-on/Roll-off vessels that differ markedly in physical structure. At 10 m GRD resolution, these sub-types are not always distinguishable from each other or from Tankers. Published results on OpenSARShip—which uses the finer Container / Bulk Carrier / Tanker split rather than the coarser AIS Cargo category—report similar confusion between those classes in the 76–89% OA range [4]. Our use of the coarser "Cargo" label likely increases confusion but reflects the realistic scenario where sub-type information is not available from AIS alone. The confusion matrix (Figure 4) shows this is almost entirely a one-way error: Tanker $\rightarrow$ Cargo misclassification is rare (7 chips, 5%), while Cargo $\rightarrow$ Tanker (22 chips, 51%) dominates. This asymmetry is physically interpretable: the Tanker class is visually coherent (long cylindrical hulls), while the Cargo class is visually diverse.

Re-ID implications and dark vessel tracking. The weak but above-chance re-ID signal at GRD resolution suggests that CROMA embeddings encode some vessel-specific signature (superstructure geometry, cargo load, trim angle) that persists across passes. In a dark vessel tracking scenario—where AIS has been disabled—a Rank-5 accuracy of 14% means the true vessel appears in the top-5 retrieved candidates roughly one in seven queries. Combined with geospatial constraints (a vessel cannot teleport between passes), this provides a useful prior for analysts. Higher-resolution sensors (TerraSAR-X at 1–3 m, Capella Space at 0.5 m) would be expected to increase re-ID performance substantially by resolving hull geometry and superstructure detail; our result establishes a lower bound at Sentinel-1 GRD resolution.

Label resolver design lessons. The signal ablation (Table III) reveals that AIS type code alone accounts for 91.5% of resolution decisions—but without the secondary signals, overall resolution would drop from 87.3% to 45.6% and HIGH-confidence assignments would collapse to just 18 vessels. The secondary signals matter not by replacing AIS type codes but by confirming or overriding them, converting MEDIUM-confidence labels into HIGH-confidence ones that are safe to train on. The length signal's counter-intuitive behavior (removing it slightly increases resolution rate while halving HIGH-confidence count) suggests a future improvement: treating length as a soft Bayesian prior rather than a hard vote, reducing its tendency to create irresolvable conflicts for vessels near the 200 m Tanker threshold.

Pipeline scalability. Because the pipeline uses only free, globally available data sources (Sentinel-1, NOAA AIS, GFW API), it can be applied to any maritime region without port-specific data collection or manual curation. The Gulf of Mexico setting—open water with low land clutter—is

particularly favorable; coastal and port scenes would require additional sea-land masking. Expanding to other ocean basins (South China Sea, Mediterranean, North Sea) would increase vessel diversity and class balance, particularly for the under-represented Fishing class (14 HIGH-confidence vessels in the current dataset).

VIII. Conclusion

We have demonstrated a zero-curation SAR ship classification and re-identification pipeline using only free Sentinel-1 GRD imagery and NOAA AIS data. A four-signal label resolver automatically assigns reliable type labels to 87.3% of vessels without any manual annotation—closing the scalability gap that has limited prior SAR ship datasets to manually curated, region-specific collections. Among three classification backbones, CROMA—pretrained natively on Sentinel-1 data—achieves 83.8% OA / 76.3% F1, competitive with DL methods trained on manually curated OpenSARShip chips (77–93% OA range [4]). A tanker re-identification experiment with ArcFace metric learning produces Rank-1 accuracy 2.5$\times$ above random chance, establishing a baseline for this novel task at C-band GRD resolution.

Data Availability

The dataset (18,342 Sentinel-1 GRD ship chips, resolved labels, and extraction scripts) is publicly available at https://doi.org/10.5281/zenodo.21000650. A secondary mirror is available at https://huggingface.co/datasets/mshaya/sar-ship-gulf.

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